Converting Invariant Cross Section to Invariant Yield

1 The two quantities

Both are built from the same azimuthally-averaged, boost-invariant differential, differing only in the normalization out front:

$$\text{invariant cross section:} \qquad E\,\frac{\mathrm{d}^3\sigma}{\mathrm{d}p^3} = \frac{1}{2\pi p_{\mathrm{T}}}\,\frac{\mathrm{d}^2\sigma}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}, \tag{1}$$

$$\text{invariant yield (per event):} \qquad E\,\frac{\mathrm{d}^3N}{\mathrm{d}p^3} = \frac{1}{2\pi p_{\mathrm{T}}}\,\frac{1}{N_{\mathrm{evt}}}\,\frac{\mathrm{d}^2N}{\mathrm{d}p_{\mathrm{T}}\,\mathrm{d}y}. \tag{2}$$

The cross section carries an area unit (e.g. $\mathrm{mb\,GeV^{-2}}c_0^3$); the per-event yield is a pure number per unit phase space ($\mathrm{GeV^{-2}}c_0^3$). The conversion is entirely a matter of what you normalize the counts by, and that is where the two collision systems part ways.

2 $p+p$: divide by the inelastic cross section

In $p+p$ the number of events sampled and the number of produced particles are both set by the integrated luminosity:

$$N_{\mathrm{evt}} = L_{\mathrm{int}}\,\sigma_{\mathrm{inel}}, \qquad \mathrm{d}^3N = L_{\mathrm{int}}\,\mathrm{d}^3\sigma. \tag{3}$$

Forming the per-event invariant yield, the luminosity cancels exactly:

$$\boxed{E\,\frac{\mathrm{d}^3N}{\mathrm{d}p^3} = \frac{1}{\sigma_{\mathrm{inel}}}\,E\,\frac{\mathrm{d}^3\sigma}{\mathrm{d}p^3}} \qquad \Longleftrightarrow \qquad E\,\frac{\mathrm{d}^3\sigma}{\mathrm{d}p^3} = \sigma_{\mathrm{inel}}\,E\,\frac{\mathrm{d}^3N}{\mathrm{d}p^3}. \tag{4}$$

So the whole conversion in $p+p$ is a single multiplicative factor: the total inelastic cross section σ_{inel}. (At $\sqrt{s} = 200\,\mathrm{GeV}$, $\sigma_{\mathrm{inel}} \approx 42\,\mathrm{mb}$.) Note L_{int} never appears in the final relation — it is a bookkeeping intermediary that drops out the moment you normalize per event.

3 $A+A$: the nuclear overlap function T_{AA}

In a heavy-ion collision there is no single σ_{inel} to divide by; the analogue of luminosity is the *nuclear overlap function* from your Glauber model,

$$\langle T_{AA}\rangle = \frac{\langle N_{\mathrm{coll}}\rangle}{\sigma_{\mathrm{inel}}^{NN}}, \tag{5}$$

which has units of inverse area and represents the binary-collision density. The binary-scaling expectation — the yield you would see if $A+A$ were just $\langle N_{\mathrm{coll}}\rangle$ independent nucleon–nucleon collisions — is

$$E\,\frac{\mathrm{d}^3N}{\mathrm{d}p^3}_{AA} \overset{\text{scaled}}{=} \langle T_{AA}\rangle\,E\,\frac{\mathrm{d}^3\sigma}{\mathrm{d}p^3}_{pp}, \tag{6}$$

i.e. the $p+p$ invariant *cross section* times $\langle T_{AA}\rangle$ gives an $A+A$ invariant *yield*. This is exactly the reference in the nuclear modification factor:

$$R_{AA}(p_{\mathrm{T}}) = \frac{E\,\dfrac{\mathrm{d}^3N}{\mathrm{d}p^3}_{AA}}{\langle T_{AA}\rangle\; E\,\dfrac{\mathrm{d}^3\sigma}{\mathrm{d}p^3}_{pp}}. \tag{7}$$

4 Why R_{CP} sidesteps the $p+p$ reference entirely

R_{CP} replaces the $p+p$ cross section with a peripheral yield, so the $E\,\dfrac{\mathrm{d}^3\sigma}{\mathrm{d}p^3}_{pp}$ reference — and the $\sigma_{\mathrm{inel}}^{NN}$ inside every T_{AA} — cancels between numerator and denominator:

$$\boxed{R_{CP}(p_{\mathrm{T}}) = \frac{\left[\,E\,\dfrac{\mathrm{d}^3N}{\mathrm{d}p^3}_{AA} \Big/ \langle N_{\mathrm{coll}}\rangle\,\right]_{\mathrm{central}}}{\left[\,E\,\dfrac{\mathrm{d}^3N}{\mathrm{d}p^3}_{AA} \Big/ \langle N_{\mathrm{coll}}\rangle\,\right]_{\mathrm{peripheral}}}} \tag{8}$$

What survives is the ratio of centrality-scaled *invariant yields*, weighted only by $\langle N_{\mathrm{coll}}\rangle$ from Glauber. You therefore never need an absolute cross-section conversion for R_{CP}: you extract the invariant yield in each centrality class directly (raw peak $\rightarrow$ $A{\cdot}\varepsilon$ correction $\rightarrow$ per-event normalization) and scale each by its own $\langle N_{\mathrm{coll}}\rangle$. The conversion factors that are hardest to pin down (σ_{inel}, absolute luminosity, the $p+p$ reference) all drop out.

5 Units summary

Quantity	Normalization	Typical unit
Invariant cross section	per unit luminosity	$\mathrm{mb\,GeV^{-2}}c_0^3$
Invariant yield $(p+p)$	$\boldsymbol{\nabla}\cdot\,\sigma_{\mathrm{inel}}$	$\mathrm{GeV^{-2}}c_0^3$ (per event)
Binary-scaled $A+A$ yield	$\times\,\langle T_{AA}\rangle$	$\mathrm{GeV^{-2}}c_0^3$ (per event)

One-line takeaway. Invariant yield and invariant cross section differ by a single normalization factor: divide by σ_{inel} in $p+p$, or multiply the $p+p$ cross section by $\langle T_{AA}\rangle$ to land the binary-scaled $A+A$ yield — and for R_{CP} that factor cancels, so you work in invariant yields throughout.